

Protein and its Structure

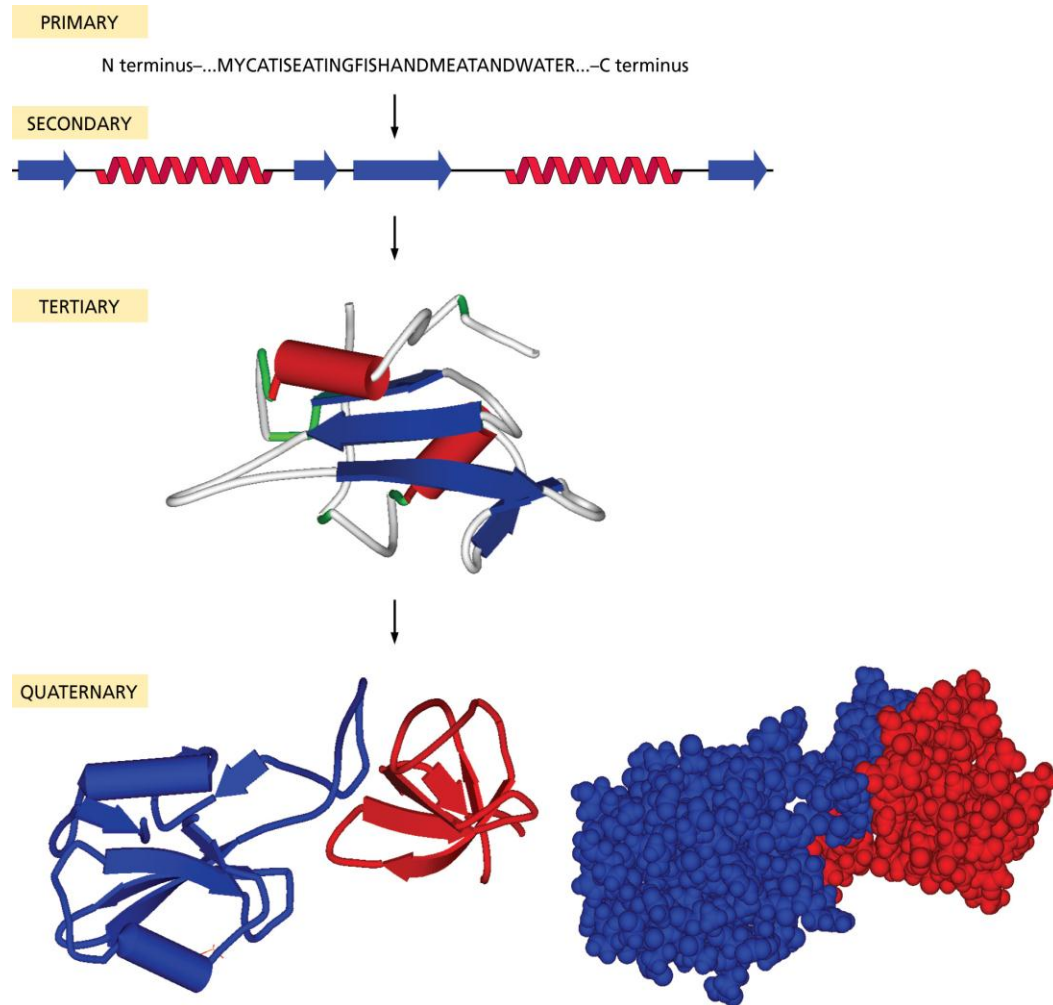
CBI 104

Chapter 2

- What is protein structure?
- What are proteins made of?
- What forces determine protein structure?
- What is protein secondary structure?
- How are protein structures determined experimentally?
- How can structures be predicted *in silico*?

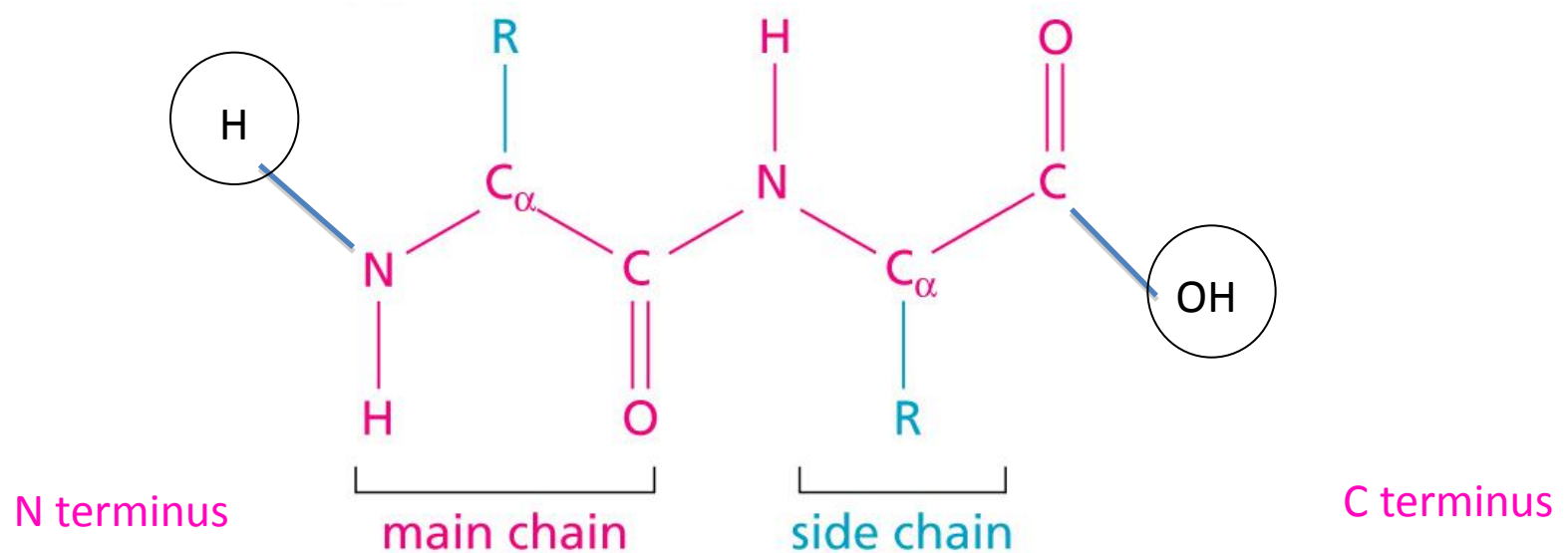
What is protein structure?

Proteins are linear polymers that fold up by themselves...mostly



What are proteins made of?

The parts of a protein



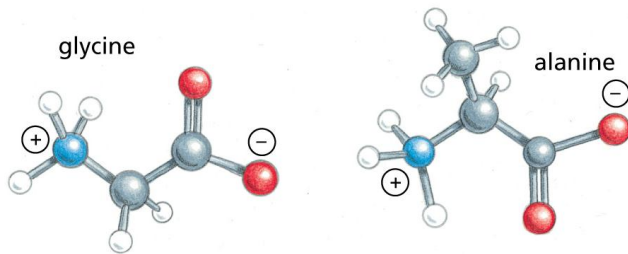
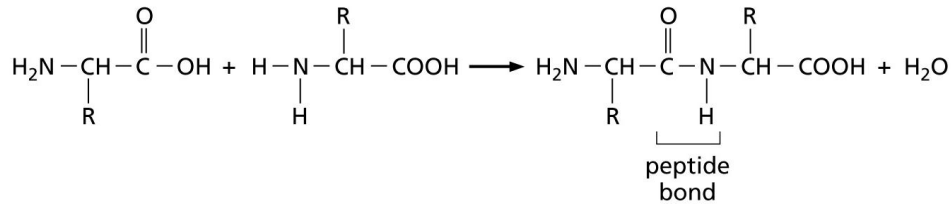
N, C, C, N, C, C... : "Backbone"

R: "side chain"

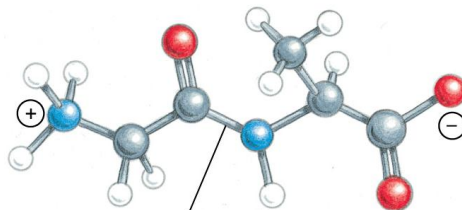
C, N, H, O, S

polymers of amino acids

Polypeptide



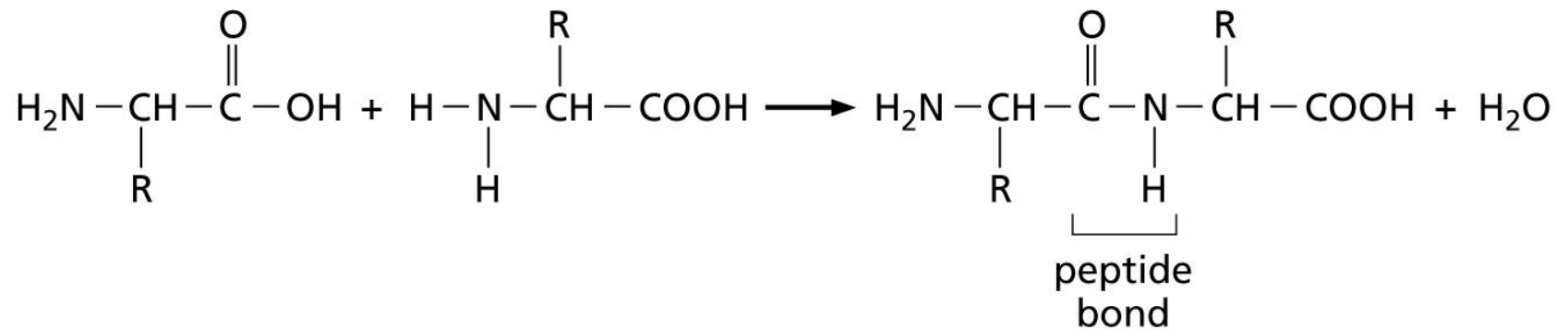
PEPTIDE BOND
FORMATION WITH
REMOVAL OF WATER



peptide bond in glycylalanine

- Bond between carbonyl carbon to amine nitrogen – **peptide bond**
- Two or more amino acids attached by a peptide bond - **polypeptide**
- In aqueous solution, C-terminus loses a proton, N-terminus picks up a proton

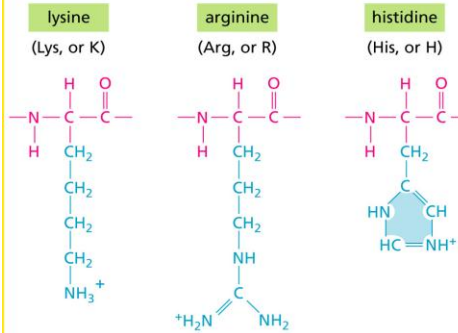
Peptide Bond



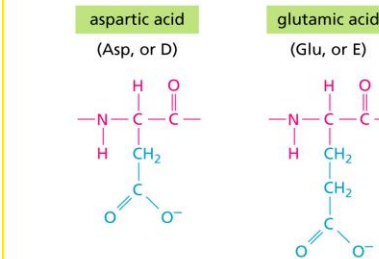
The amino acids

- Can be grouped by properties in many ways according to the chemical and physical properties (e.g. size) of the side chain.
- One grouping based on chemical properties:
 - Basic: proton acceptors
 - Acidic: proton donors
 - Uncharged polar: have polar groups like CONH_2 or CH_2OH
 - Nonpolar: tend to be hydrophobic
 - Weird: proline links to the N in the main chain
 - Strong: Cysteine can make "disulphide bridges"

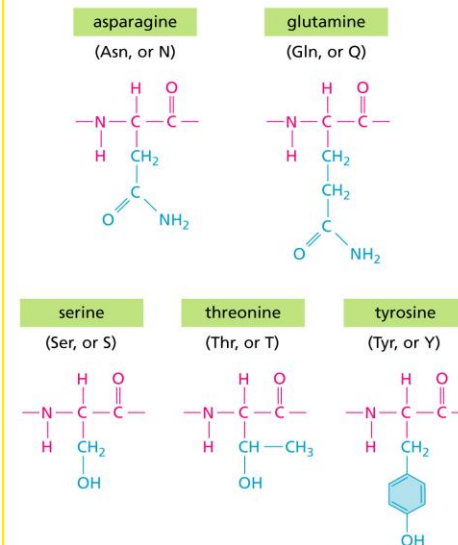
BASIC SIDE CHAINS



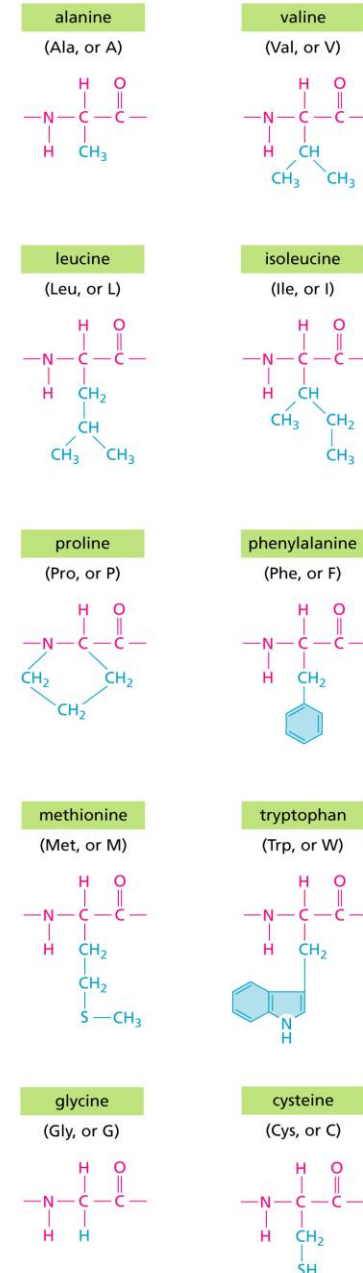
ACIDIC SIDE CHAINS



UNCHARGED POLAR SIDE CHAINS



NONPOLAR SIDE CHAINS

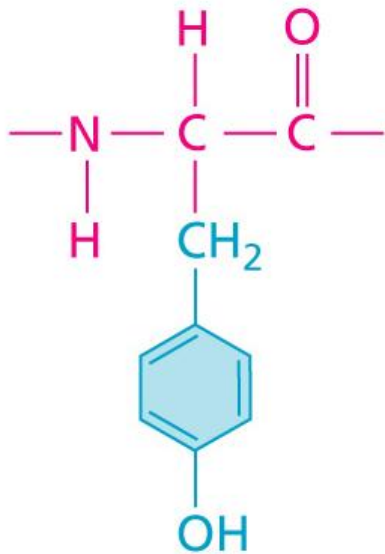


Simplest Side Group: hydrogen

Glycine

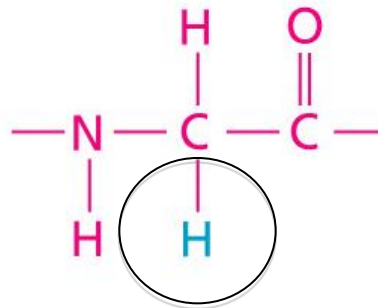
tyrosine

(Tyr, or Y)



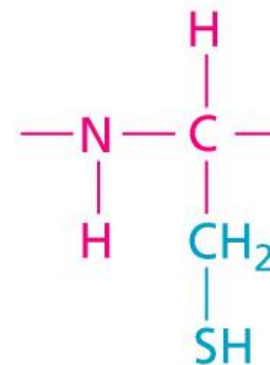
glycine

(Gly, or G)



cysteine

(Cys, or C)

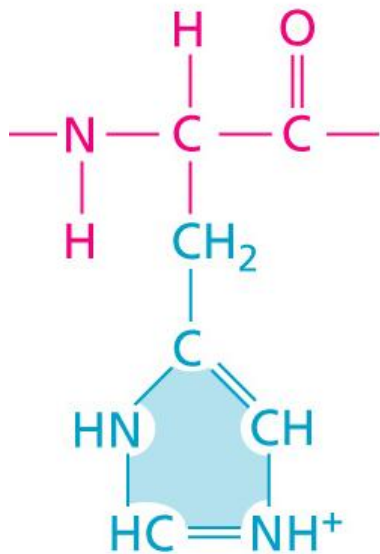


All others start with a methyl group

Simplest is Alanine

histidine

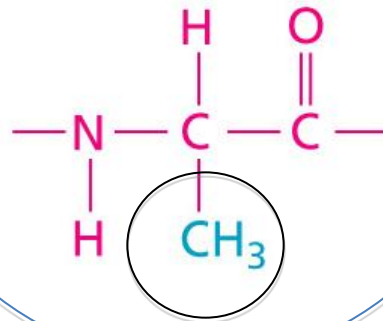
(His, or H)



NONPOLAR SIDE CHAINS

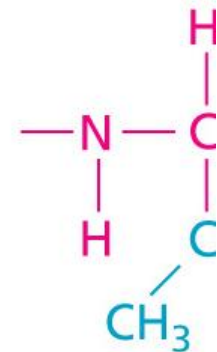
alanine

(Ala, or A)



vali

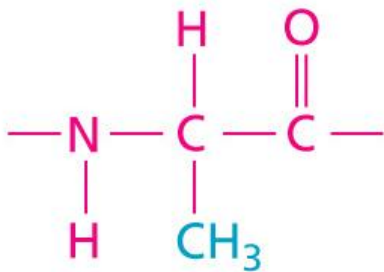
(Val, or V)



Add phenyl group to Alanine: Phenylalanine

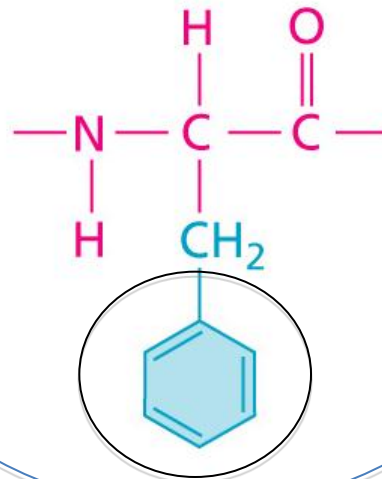
alanine

(Ala, or A)



phenylalanine

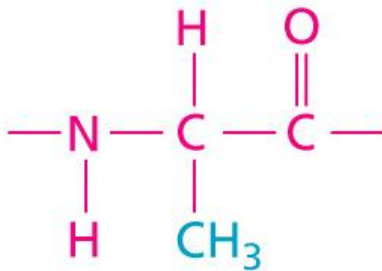
(Phe, or F)



Add hydroxyl group to Alanine: Serine

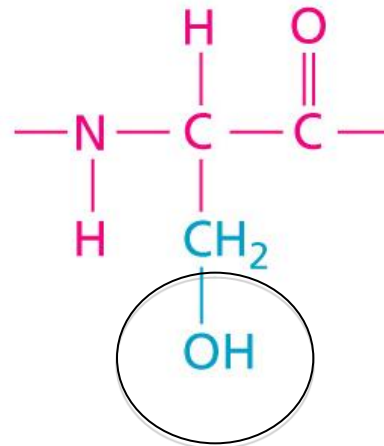
alanine

(Ala, or A)



serine

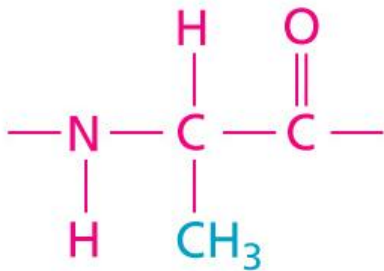
(Ser, or S)



Add SH group to Alanine: Cysteine

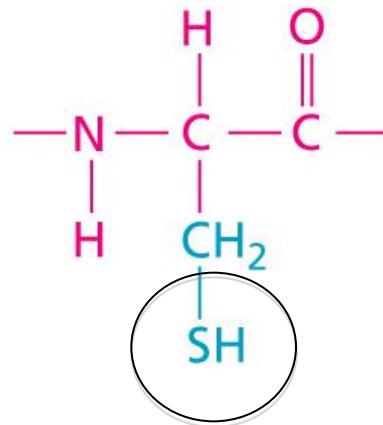
alanine

(Ala, or A)



cysteine

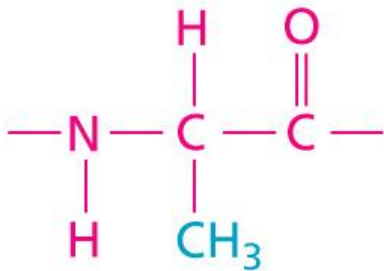
(Cys, or C)



Add carboxyl group to Alanine: Aspartic Acid

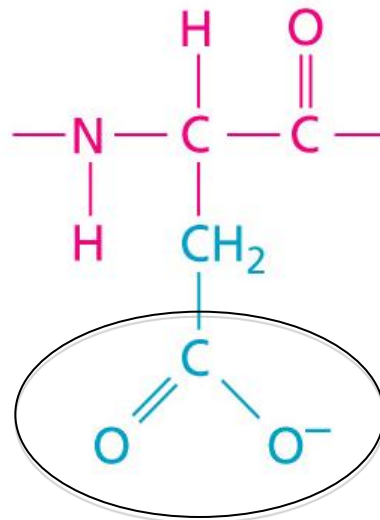
alanine

(Ala, or A)



aspartic acid

(Asp, or D)



How are proteins formed?

- The primary information for protein is encoded by the corresponding gene
- Transcription of the gene – mRNA; translation of mRNA – protein
- Ribosome reads mRNA in nucleotide triplets [codons] coding for specific amino acid
- Each triplet is specific for a particular amino acid – GENETIC CODE

Primary Structure

- The sequence of the amino acids in the protein
- Has a N-terminus (NH_2) on the left to the C-terminus ($-\text{COOH}$) on the right
- Information about the folding pattern or the final active form of the protein
 - *sequence* specifies *conformation* specifies *function*
- Used for prediction of secondary and tertiary structures

What forces determine protein structure?

Minimum free energy

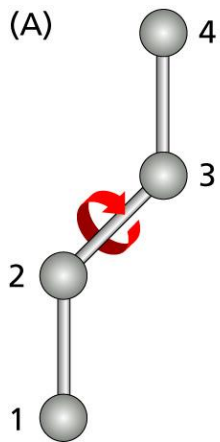
- Proteins tend to fold naturally to the state of minimum free energy (Christian Anfinsen)
- This state is determined by forces due to interactions among the residues
- Proteins usually fold in an aqueous environment, so interactions with water molecules are key
- Some proteins fold in membranes, so interactions with lipids are important

Atomic Bonds

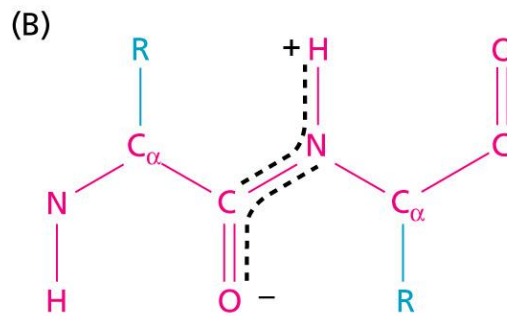
- Covalent bonds – strong
 - Single bonds can usually rotate freely
 - Double bonds are rigid
- Hydrogen bonds – weak
 - Oxygen and Nitrogen share a proton (Hydrogen)
- Van der Waals forces – weaker still

Planar Peptide bond

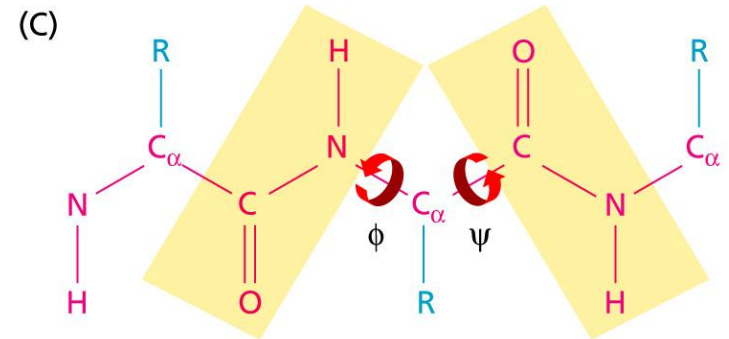
Flexible C-alpha bonds



Single bonds rotate



Resonance makes peptide bonds planar



The C-alpha bonds have two free rotation angles: phi and psi

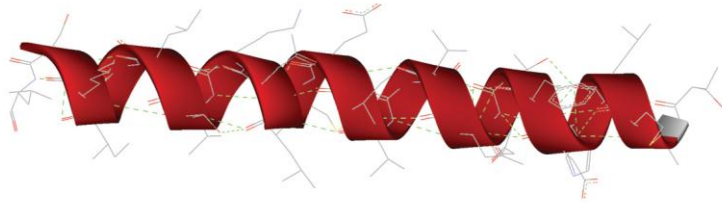
Secondary Structures

Certain repetitive structures are energetically favorable

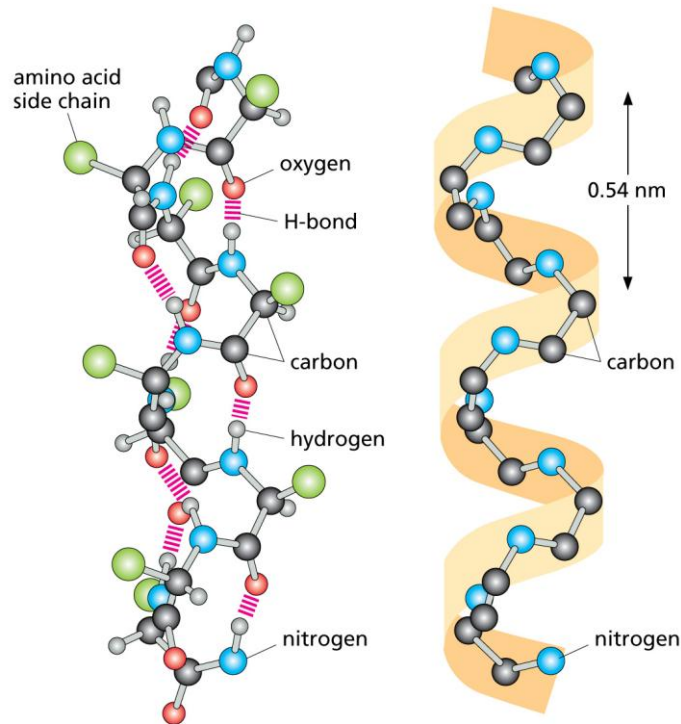
- hydrogen bonds formed among residues
- do not encounter lots of steric hindrances
- They occur over and over again in natural proteins
- Some combinations of secondary structures are so common they are called “folds” (e.g., the SCOP database of protein folds)

Alpha Helix

(A)

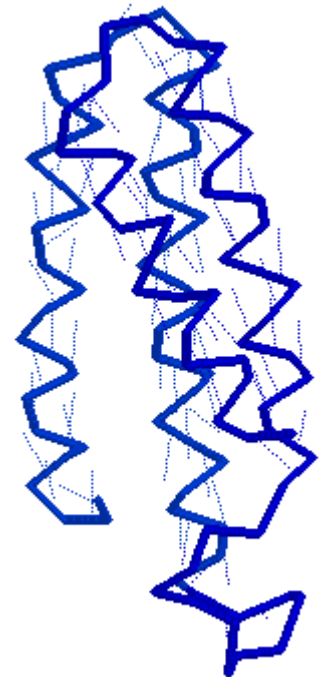
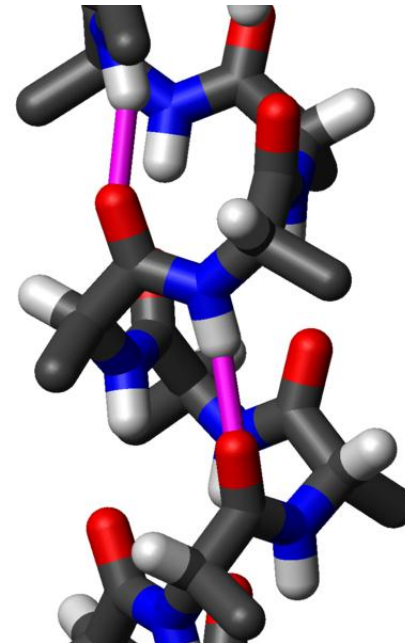


(B)

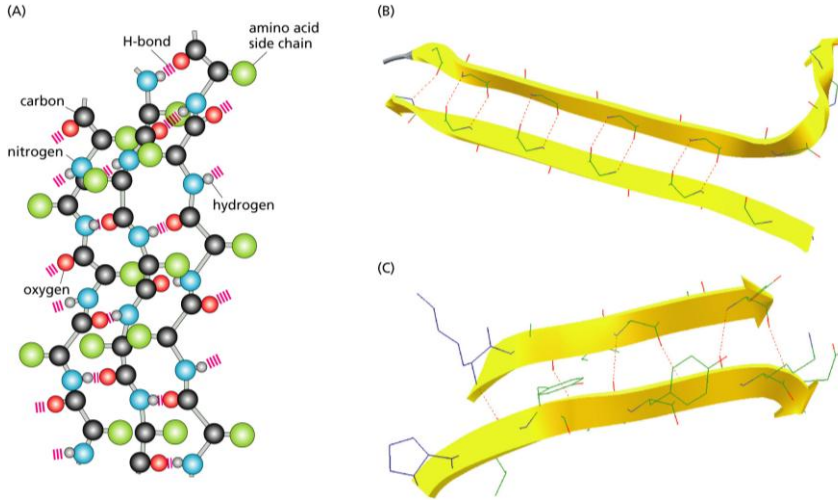


alanine, glutamate, and leucine tend to be present in α helices

- Characterised by H bonds along the chain, almost co-axial
- 3.6 amino acid (residues) per turn
- $O(i)$ hydrogen bonds to $N(i+4)$



Beta Sheet



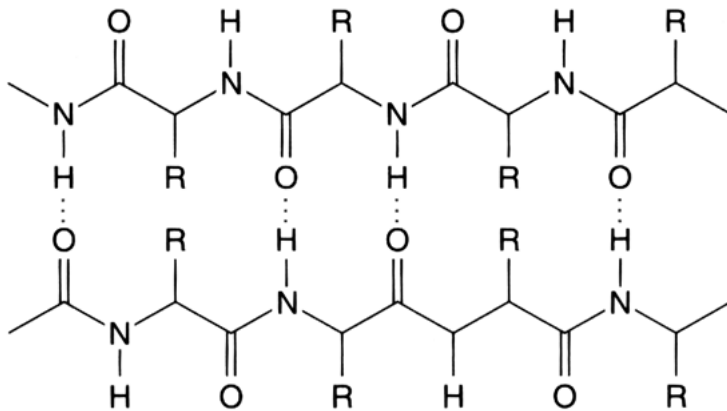
A. Three strands shown

B. Anti-parallel sheet

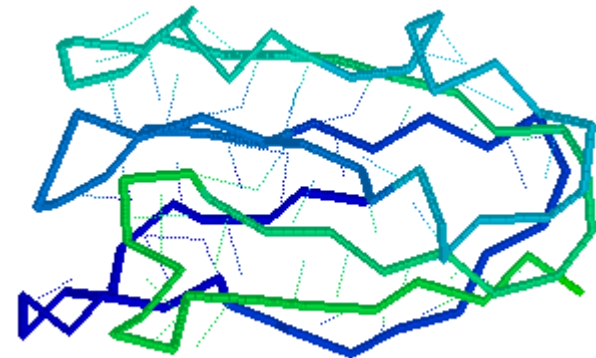
C. Parallel sheet

Characterised by H bonds crossing between chains

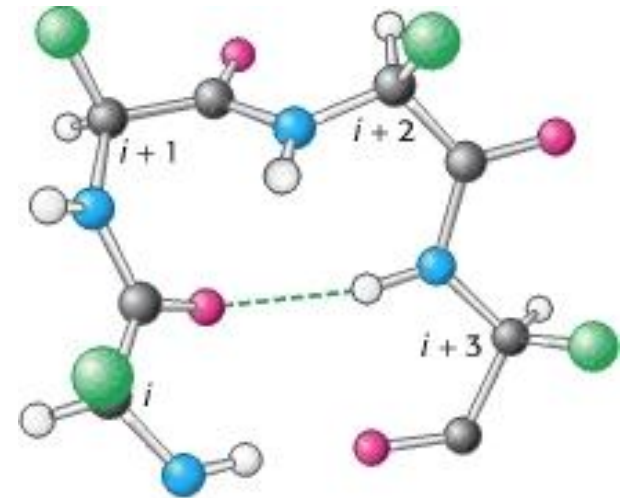
Sheets are usually curved and can even form barrels



Valine, isoleucine tend to be present in β strands



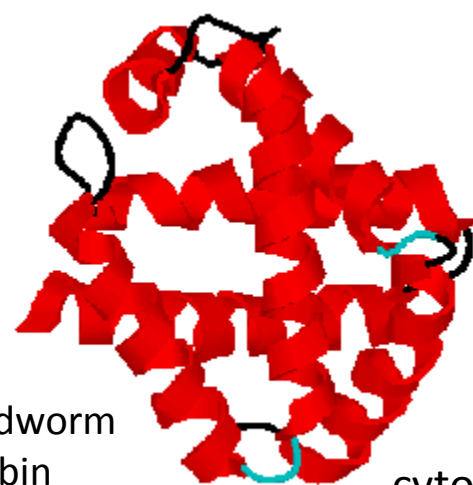
- The CO group of residue i of the polypeptide chain is hydrogen bonded to the NH group of residue $i + 3$ to stabilize the turn



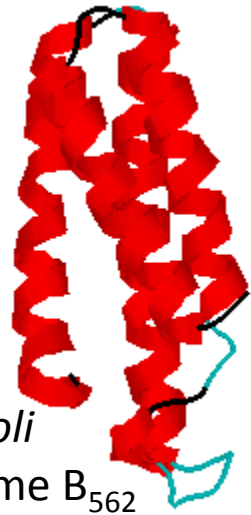
Wheat germ agglutinin - 16 turns
(cyan), 16 disulfide bridges (brown)



Marine Bloodworm
hemoglobin



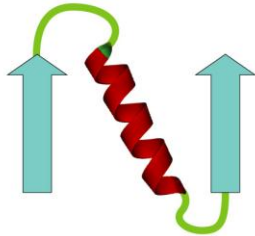
E. coli
cytochrome B₅₆₂



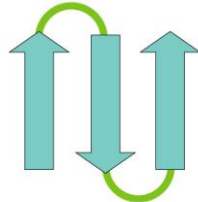
Supersecondary Structure

Adjacent elements of secondary structure form structures bigger than secondary structure smaller than subunits or motifs

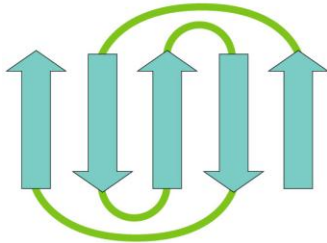
(A)



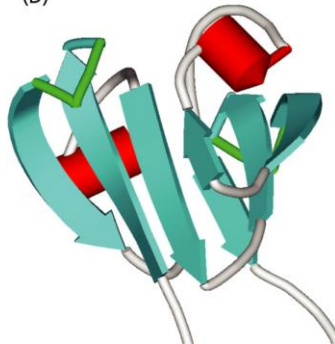
(B)



(C)



(D)



A: beta-alpha-beta

B: beta-meander

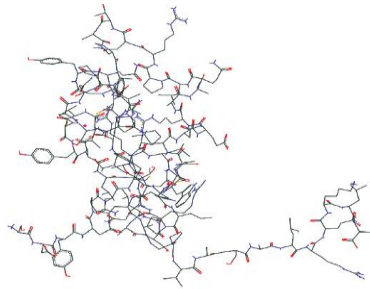
C: Greek-key

D: Greek-key

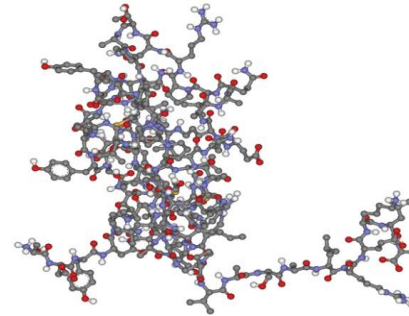
E: helix-turn-helix

The context is often crucial in determining the conformational outcome in protein structure

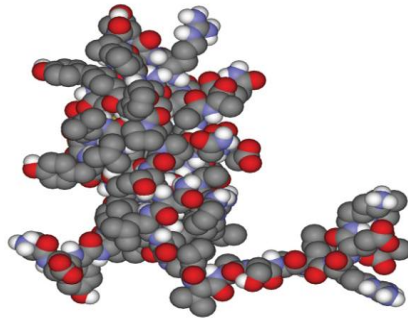
Tertiary Structure



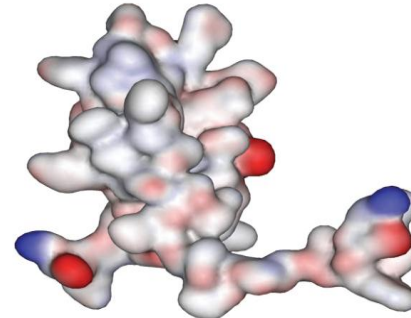
wire-frame



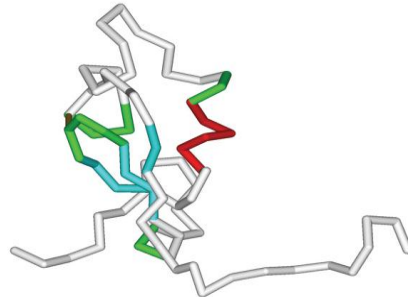
ball and stick



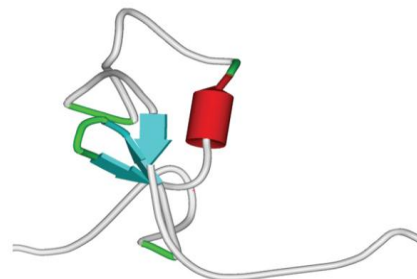
space-filling



surface



C α representation



α/β schematic

Folds

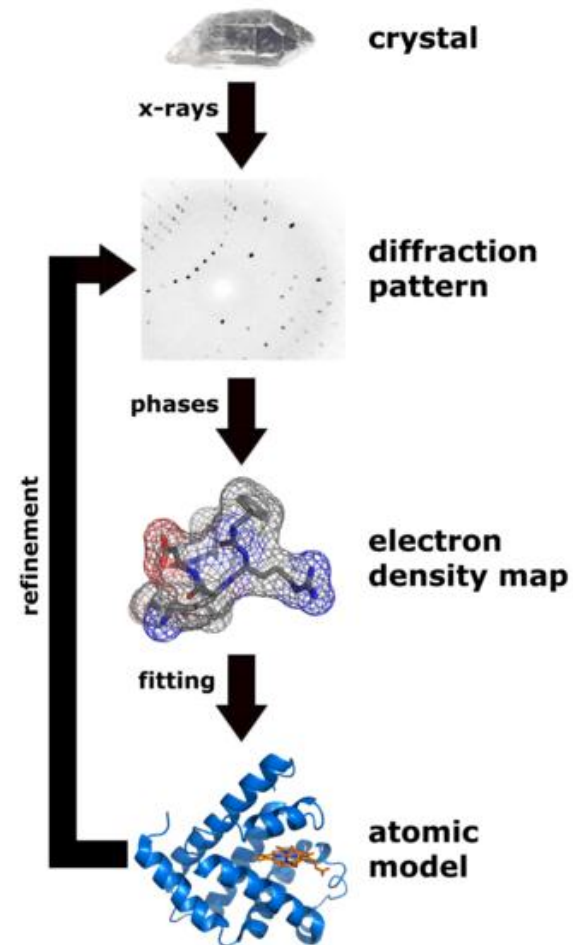
- Folds are way to classify proteins by tertiary structure
- SCOP: Structural Classification of Proteins

Class	Number of folds	Number of superfamilies	Number of families
All alpha proteins	259	459	772
All beta proteins	165	331	679
Alpha and beta proteins (a/b)	141	232	736
Alpha and beta proteins (a+b)	334	488	897
Multi-domain proteins	53	53	74
Membrane and cell surface proteins	50	92	104
Small proteins	85	122	202
Total	1086	1777	3464

How is protein structure
determined experimentally?

X-ray crystallography

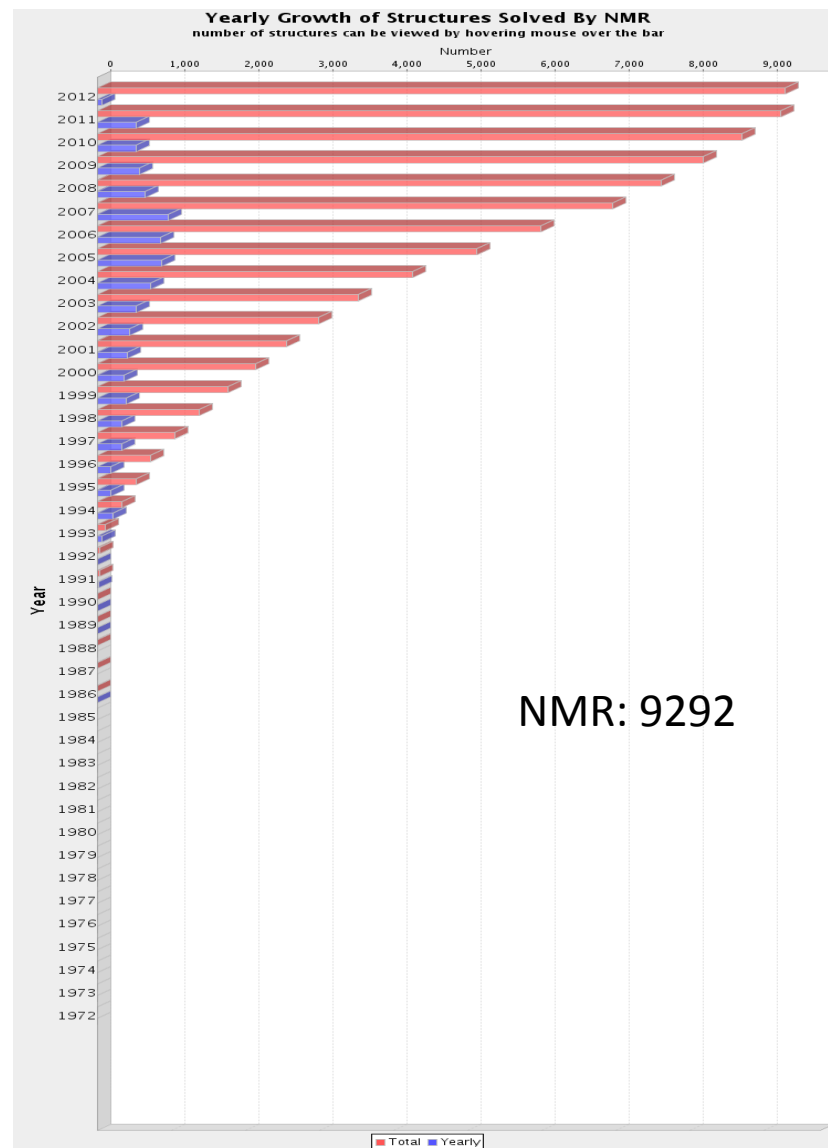
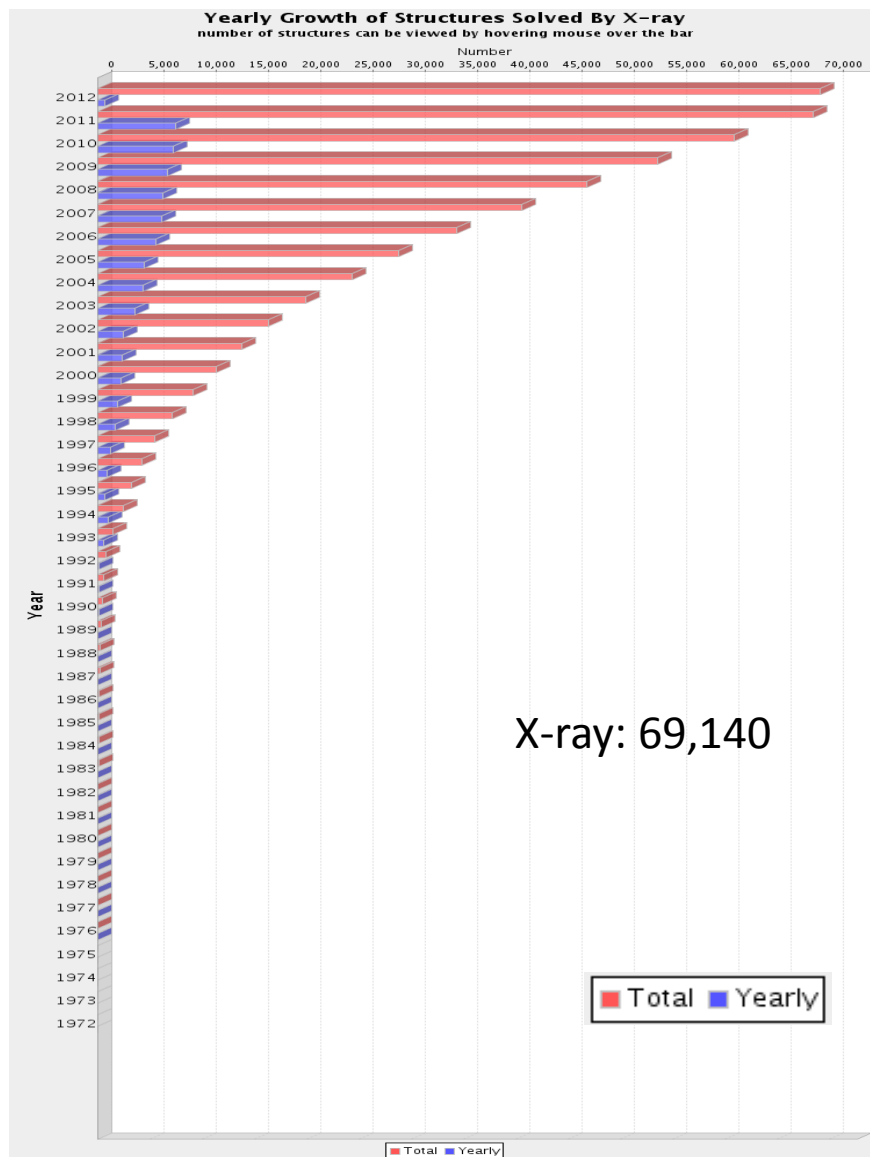
- Needs crystallized proteins
- Hard to get crystals
- Very tough for hydrophobic (e.g. transmembrane) proteins
- Better accuracy than NMR
- Expensive:
\$100,000/protein



NMR spectroscopy

- Protons resonate at a frequency that depends on their chemical environment
- This can be used to predict structure
- Does not require crystallization; protein may be in solution
- Lower resolution than X-ray crystallography

Protein DataBank (PDB)



How can protein structure be
predicted *in silico*?

Tertiary structure prediction is still too hard

- Comparative modeling
 - Uses sequence alignment to protein of known structure
 - E.g., Modeller
- *Ab initio* modeling
 - Uses primary sequence only
 - E.g., Rosetta



Rosetta prediction

Secondary Structure Prediction

- Much simpler to predict a small set of classes than to predict 3-D coordinates of atoms
- Amino acids have different propensities for alpha helices, turns and beta sheets
 - A,D,M,I,E,L – form α helices; G, P disrupt
- Homology can also be used since fold is more conserved than sequence

- www.ncbi.nlm.nih.gov
- www.rscb.org [wwPDB]
- <http://www.expasy.ch>